

# Tone burst generator

## Introduction:

A tone burst generator is a circuit that produces a burst of sound at a specific frequency for a short duration. It can be used in applications like audio testing, signaling, or as a tone generator for alarms or communication systems. The 555 timer IC is a popular choice for building tone burst generators because of its versatility, ease of use, and low cost.

## Objective:

Creating a straightforward and efficient tone burst generator utilizing the 555 timer IC, that acts as a flexible tool capable of producing audible tones suitable for a range of applications, including alarms, notifications, or signaling mechanisms.

## Equipment Required:

1. Resistors (1K, 47K, 10K, 4.7K, 100K, 100  $\Omega$ )
2. 555 Timer IC
3. Capacitors (0.01 $\mu$ F(2p), 100  $\mu$ F)
4. 9V battery
5. Veroboard
6. Connecting Wires
7. Speaker(16ohm)
8. NPN transistor (2N2222)
9. Switch

## Circuit Diagram:

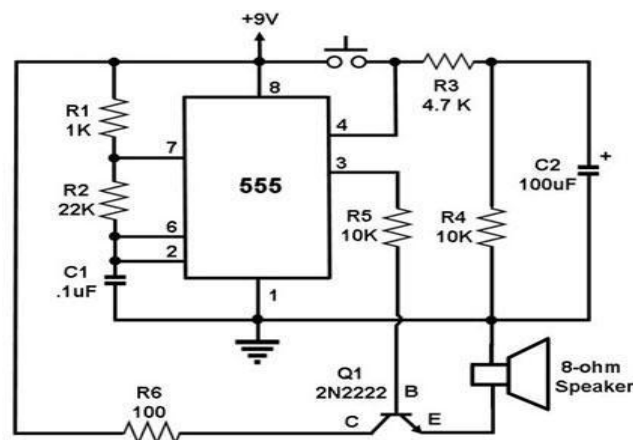


Figure: Tone Burst Generator Circuit

## Working Operation:

A tone burst generator using 555 can be explained as follows:

The 555 timer is configured in astable mode to generate a continuous square wave at a specific frequency. This output is used to create the tone. To produce a tone burst (a controlled series of pulses), the 555 is controlled by another circuit or mechanism that limits its operation to a specified duration.

The circuit includes components like resistors and capacitors to set the frequency of the tone, while the duration of the burst is controlled by an external trigger or a second 555 timer operating in monostable mode. The monostable timer determines the on-time of the astable circuit by using its RESET pin to enable or disable the tone generation. Once the set duration ends, the tone stops, completing the burst.

## Block Diagram:

Here's a simplified block diagram of a tone burst generator using a 555 timer:



In this diagram, the 555 timer is configured in astable mode to generate a square wave at a specific frequency. Control logic is used to trigger the timer to produce tone bursts. The power supply provides necessary voltage for the entire circuit.

## Application of the project:

A tone burst generator using the 555 timer is typically used to generate a series of audio tone pulses for various applications, such as signaling, alarms, or audio testing.

In this setup, the 555 timer is configured in astable multivibrator mode, generating a series of square wave pulses. These pulses can be fed into a speaker or audio output device to produce a tone burst at a specified frequency. The duration of the tone burst is controlled by the timing components (resistors and capacitors) connected to the 555, allowing adjustment of the pulse width and interval between bursts.

For instance, a tone burst generator using the 555 can be used in a medical alert system, producing short, periodic tones to indicate status changes or alarms. Additionally, it can be

applied in electronic testing for generating short bursts of sound for troubleshooting audio equipment.

**Speaker Testing:** Generating tone bursts for checking speaker response.

**Audio Signal Testing:** Used in circuits where a specific burst of tone is needed for system testing or diagnostics.

**Alarm or Indicator System:** Used in applications where periodic or short alerts are required, like in timers or signaling systems.

### Cost Calculation:

1. Resistors (1K, 47K, 10K, 4.7K, 100K, 100  $\Omega$ ).. ..... 30 tk

2. NE555 Timer IC..... 20 tk

3. Capacitors (0.01 $\mu$ F(2p), 100  $\mu$ F) ..... 10 tk

4. 9V battery.....60 tk

5. Veroboard.....30 tk

6. Connecting Wires.....10 tk

7. Speaker (8 ohm) .....20 tk

8. NPN transistor (2N2222) ....2tk

9. Switch.....5tk

.....

total cost=187 tk

### Complex Engineering Problems (P):

P1 - Depth of Knowledge

P2 - Conflicting Requirements

P3 - Depth of Analysis

P4 - Familiarity of Issues

P5 - Extent of Applicable Codes

P6 - Extent of Stakeholders

P7 – Interdependence

## Knowledge Profile (K):

K1 - Natural Sciences  
 K2 - Mathematics  
 K3 - Engineering Fundamentals  
 K4 - Specialist Knowledge  
 K5 - Engineering Design  
 K6 - Engineering Practice

## Complex Engineering Problem Condition:

According to BAETE Manuel 2019, our project satisfies the knowledge profile of K3 (Engineering Fundamentals), K4 (Specialist Knowledge), K5 (Engineering Design), and K6 (Engineering Practice). So, this project satisfies complex engineering problem P1 (Depth of Knowledge). This project also satisfies P2 (Range of Conflicting Requirements).

## Justification of Complex Engineering Problem:

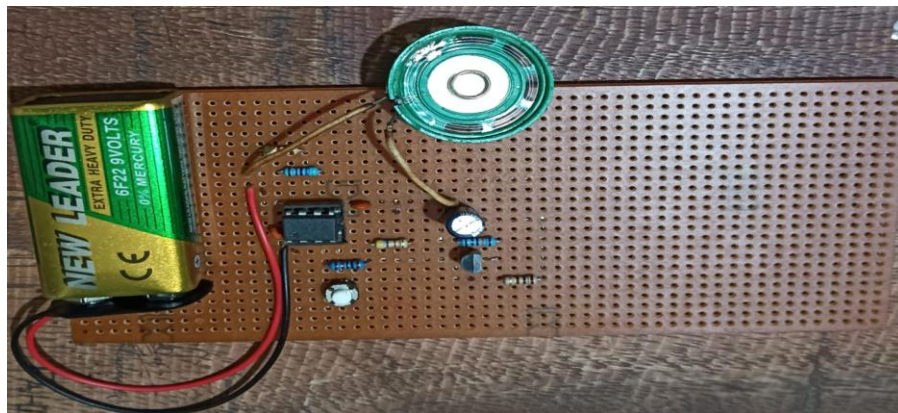
P1: This project satisfies problem 1 because it satisfies K3, K4, K5, and K6.

P2: This project satisfies problem 2 because we maintain the quality of this project and also use fewer components, and the cost is minimal. Therefore, it's a complex engineering problem.

## Conclusion:

In conclusion, the NE555 timer is a simple, cost-effective solution for generating tone bursts. By configuring it in astable mode, it can produce a square wave tone, while additional circuitry can control the burst duration and pause between tones. This makes it ideal for applications in audio testing, alarms, and communication systems.

## Picture of the project:



**Project Video Link:** <https://youtube.com/shorts/0nkiljsExag?feature=share>